A robot that acts to know

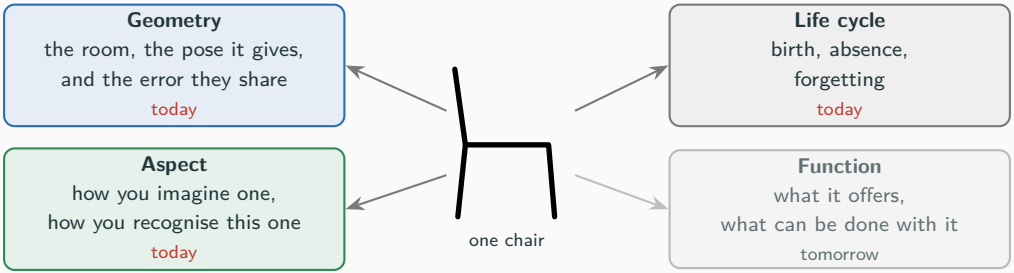
Day 3 — Concepts: geometry, life cycle, aspect

Course built on the doctoral thesis of Noé Zapata

Wednesday 2 December

Montevideo, 30 November – 4 December 2026

Yesterday: a concept has dimensions. Today: three of them

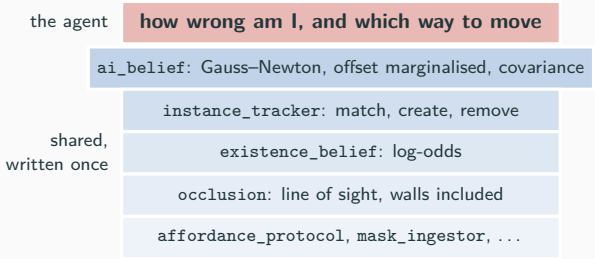


First, the thing that answers for all of them: the concept agent.

source: the dimensions are this course's way of laying out the thesis

What a concept agent is

An agent writes two things

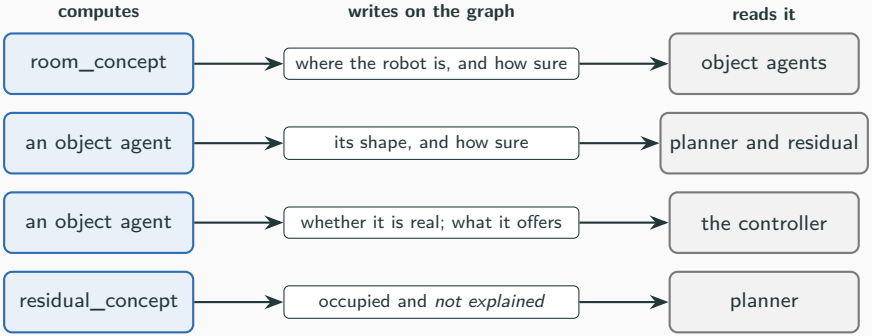


A residual and its Jacobian. All the Bayesian mathematics lives in the engine, once.

This is why a roster entry can be half a page: bottle needs little, table and refrigerator fill the template, and that is a fact about the objects.

source: concept_agents.tex l.534–537, l.136–138, l.1196–1203

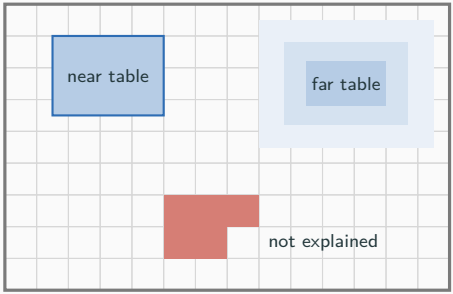
Each publishes what only it can compute



Nobody asks anybody. Everything an agent knows about another's work is on the graph.

source: concept_agents.tex l.1209–1216, l.1533–1544

residual_concept: whatever nobody else explains



a sure table claims its cells tightly; an unsure one loosely

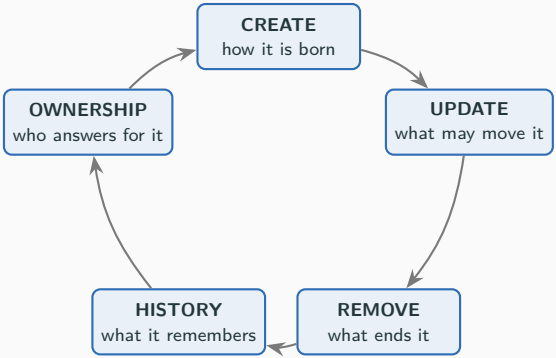
A grid of what the sensors hit, minus what every other agent claims to explain. What is left is what the robot must avoid.

Nothing has to be recognised for the robot not to hit it.

No safety margin anywhere: each claim is as wide as the claimant's own published uncertainty. That is why the others must publish it.

source: concept_agents.tex l.1497–1538

Every agent answers to a common contract



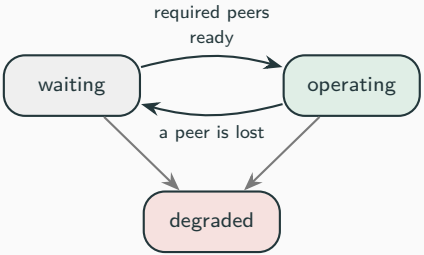
Not paperwork: this is what anything that integrates noisy fragments has to do.

A rule written in a contract is a rule a new agent cannot forget.

Next: UPDATE is geometry; CREATE and REMOVE are the life cycle.

source: concept_agents.tex l.169–180; CONCEPT_AGENT_LIFECYCLE.md l.7–13

The factorisation is only true while its owners are alive



Specified in TLA$^+$ and model-checked: 8 safety and 4 liveness properties, no counterexample.

A process crashes and its node stays. It looks like any other data, and nobody is keeping it true any more.

A peer that is present is *running*, which is not the same as *able to serve*. Heartbeats are ordinary graph nodes.

source: cortex.tex l.732–741, l.757–775, l.780–785

Geometry

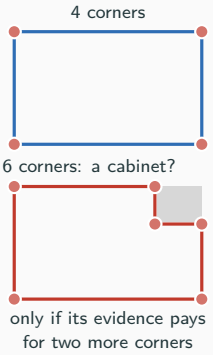
The room is the first belief born

How many walls? Nobody knows, so the room's shape is itself a choice, and each corner has a price.

One corner placed to 3 cm, in a room 10 m across:

$$\ln(10/0.03) \approx 5.8 \text{ nats.}$$

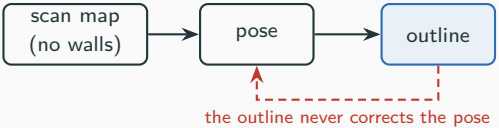
A corner is kept only if the evidence pays for it, the trade an object will face at birth. The shortest description of the room and the lowest free energy are, here, one rule.



source: layout_acquisition.tex l.56–66, l.128–146

Two regimes, and a loop that is never formed

acquisition: the room is a variable



operating: the room is a constant

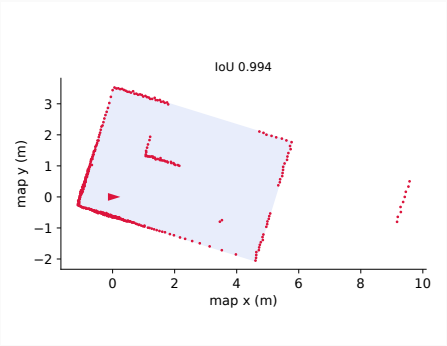


plain SLAM keeps running as a *shadow*:
a pose that measures how wrong the room might be

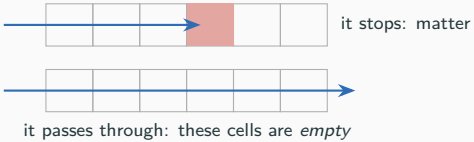
A wrong wall would drag the pose, and the dragged pose would confirm the wrong wall, so the loop is never closed. The price: a constant cannot be doubted by the localiser that leans on it.

source: layout_acquisition.tex l.162–199

What tells a wall from a table: free space



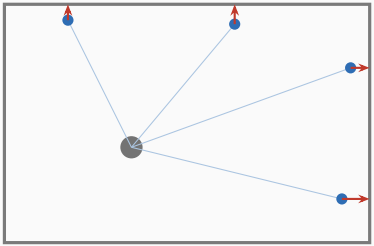
The table's south face earns **82 nats** as a candidate wall, against a price of 7.6. Evidence for a surface cannot keep furniture out of the room.



Only a beam that passes can say a cell is empty. Free floor beyond an edge is evidence against that edge.

source: lab 3, exercise 11; layout_acquisition.tex l.507–526

The room supplies the likelihood



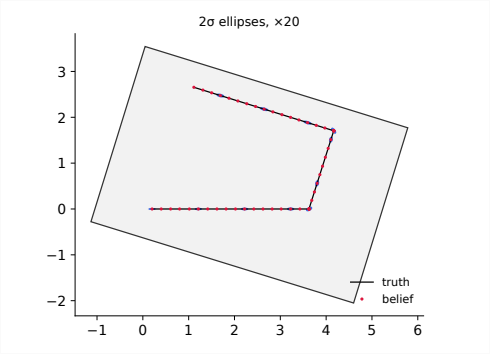
each return is pulled onto its nearest wall

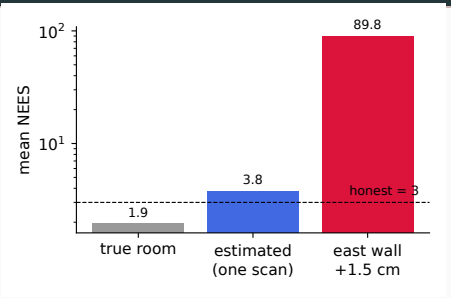
Where would I have to stand for this scan to land on these walls?

- The distance from each return to its wall is the prediction error. No expected scan is ever drawn.
- More points in a scan do not make the belief more certain: their errors are not independent.
- Range and incidence weigh each point: precisions, not gates.

source: room_concept.tex l.243–307

Localising against the room you just built



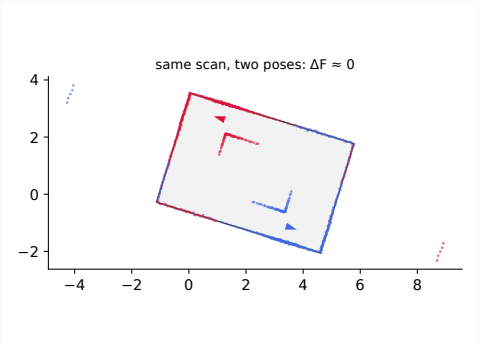


A wall **1.5 cm** out of place makes the localiser confidently wrong: NEES 90 where 3 is honest.

In the operating regime the polygon is a constant, so its error is shared by every scan, for ever.

source: lab 3, exercise 13

Symmetry: where descent cannot help



A rectangle looks the same after a half-turn: the same scan fits both poses with *identical* free energy, and no downhill path joins them.

They are separate hypotheses, so the agent enumerates them and chooses: a discrete search, then local refinement.

This is one of the few places where a threshold is right: a threshold is defensible exactly when the alternative cannot be expressed as a covariance.

source: room_concept.tex l.1869–1927

Association: the mistake the free energy cannot see



a poorly known object accepts detections from further away

Each detection goes to the object it most likely belongs to: one best guess, closest first.

A guess has no uncertainty to pay for, so it adds nothing to F. A wrong match is invisible to the quantity that would have registered it.

The repair comes later and from elsewhere: merging, and the existence belief.

source: concept_agents.tex l.483–486, l.859–889

The common-mode ceiling



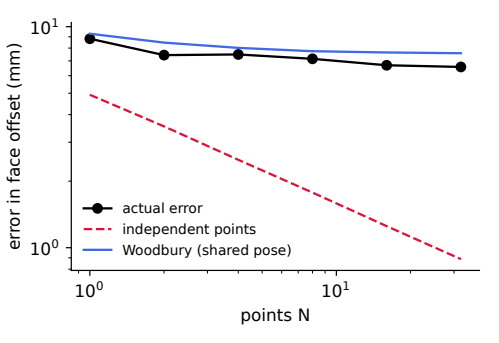
The same arrow as yesterday, now on a thousand points.

Averaging points removes their small independent jitter. It cannot touch the arrow.

A denser mask eventually stops helping: the limit is set by the robot's pose, not by the number of pixels. And the data cannot tell a displaced object from a displaced robot.

source: concept_agents.tex l.1048–1090

The ceiling, measured in the toy



The robot maps the table's face from a *dead-reckoned* pose.

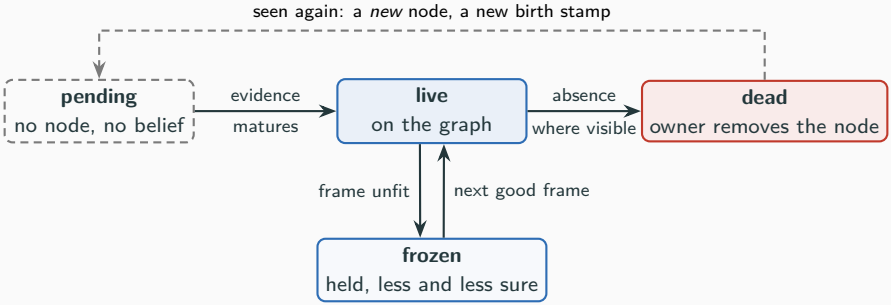
- Treating points as independent claims under 1 mm at $N = 32$.
- The real error stalls near 7 mm.
- Woodbury predicts it to within about 15% at every N.

Averaging removes independent error. It does not remove shared error.

source: lab 3, exercise 15

Life cycle

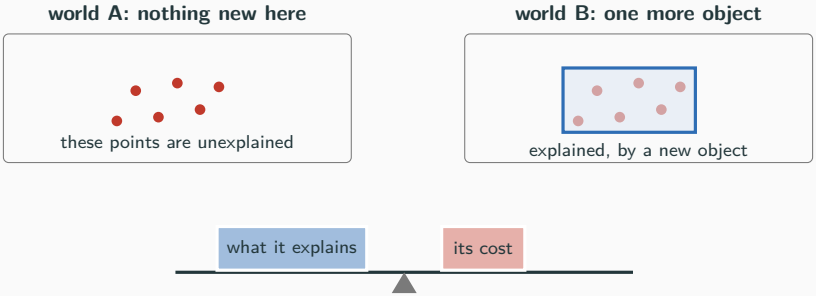
The life of an instance



Pending exists so that a wrong hypothesis costs nothing: nobody has read it yet.
Frozen suppresses an instance's *intake*, not the instance.

source: agents_life_cycle.tex l.379–423

Birth is a comparison between two worlds

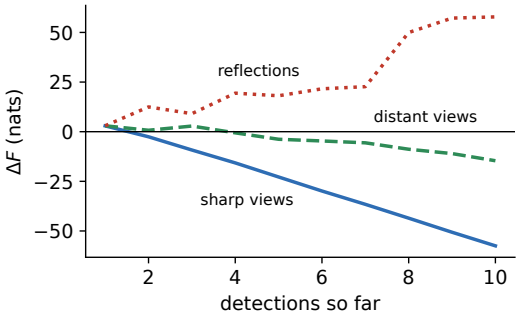


The object is born exactly when what it explains outweighs what it costs, both in nats: the room's corners faced the same trade.

The cost is not a penalty added on top: every live object already pays it.

source: agents_life_cycle.tex l.594–622

A streak cannot read evidence



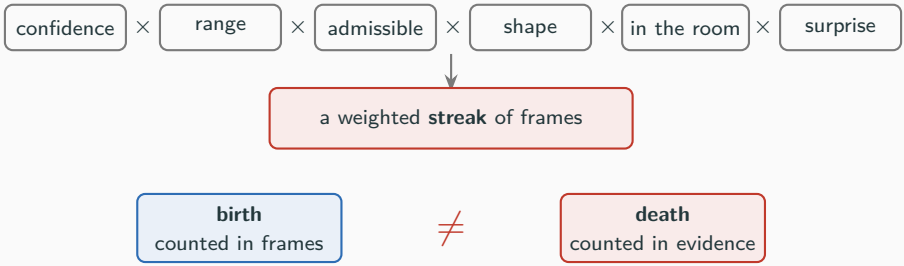
300 runs	streak of 3	$\Delta F < 0$
2 sharp views	0%	98%
10 reflections	94%	0%

A streak counts frames. ΔF reads what each frame says: how sharp it was, and how well the frames agree.

Two sharp views are worth more than ten reflections, and only nats can say so.

source: lab 3, exercises 1–2

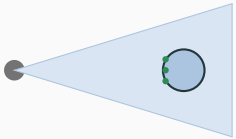
What the robot actually computes: a threshold that should not be one



Each factor is sensible, but the units are gone: birth can be traded against nothing else, so a whole subsystem exists to undo births that should not have happened. The proper comparison is within reach; what is missing is plumbing, not theory.

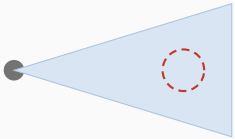
source: agents_life_cycle.tex l.629–706, l.1276–1282

A miss is evidence only where the object would have been seen



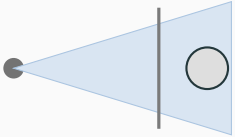
occupancy: a return

confirms



predicted visible, absent

removes



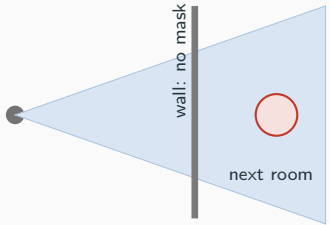
occluded or out of view

holds

The whole stage lies between the middle panel and the right one. Existence is one running number, evidence for and against simply adds to it, and it never reaches certainty, so it can always recover.

source: agents_life_cycle.tex l.887–919

A factor whose argument list was wrong



projects into view, lands on unmasked pixels, finds no mask, votes for its own removal

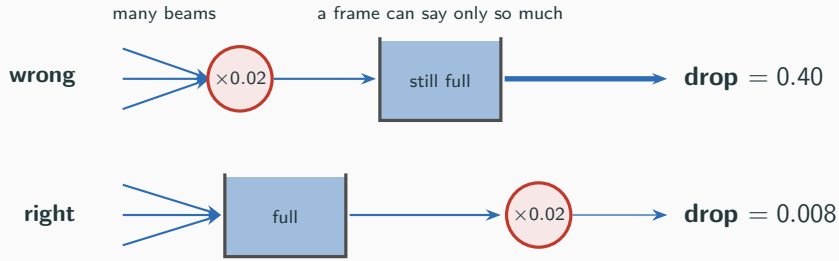
A detector has no class called “wall”, so without the room’s outline every wall is transparent. Line of sight must include the room.

Measured: the same defect, three agents

Refrigerator, then door, then table and cabinet. Each copy was consistent with itself, so no per-agent test could find it.

source: agents_life_cycle.tex l.921–951; CONCEPT_AGENT_LIFECYCLE.md l.7–11

Weight absence by how visible it was, after the bucket

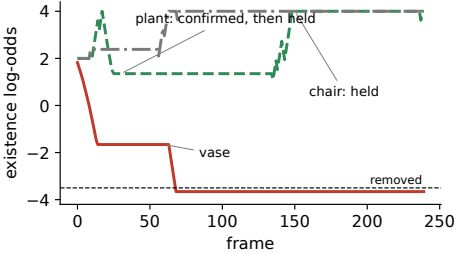


A view in which a present object would be only 2% resolvable.

Turned down before the bucket, the tap does nothing: the bucket fills anyway.

source: agents_life_cycle.tex l.953–973; numbers: lab 3, exercise 4

Four rules, one tour

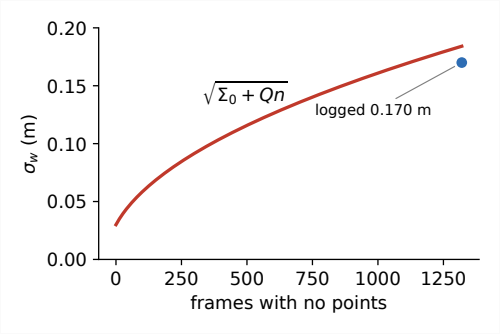


rule	removes (frame)
evidence	vase (68)
no walls in line of sight	vase (68), chair (174)
weight the count	vase (13)
miss counter	everything by frame 44

Never resolvable? No decay constant. The instance is routed to a **verification pull**: uncertainty that cannot be resolved from here is a reason to move.

source: lab 3, exercise 4; agents_life_cycle.tex l.975–985

A constant Q forgets on the agent's clock



Every frame, seen or not, a fixed amount of doubt is added.

- A table known to 3 cm loses half its precision in **3.6 s**.
- It is back at the prior after about **5.9 minutes**, whether it was tracked for thirty seconds or for three hours.

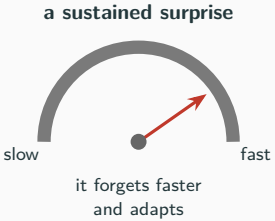
Measured: a real table, 1320 blind cycles

Predicted error 0.184 m; logged 0.170 m. The next sparse mask re-cut the table.

source: agents_life_cycle.tex l.1007–1043

The forgetting rate becomes something the object infers



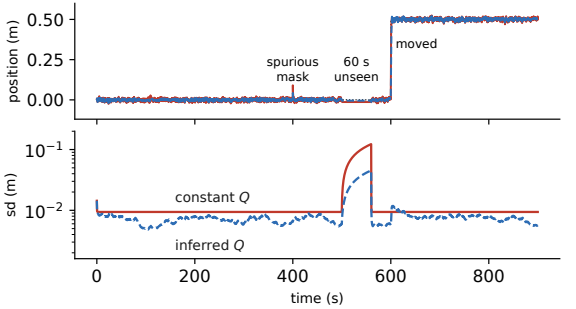


Each object turns its own dial, by how surprised it has been against how surprised it expected to be. One outrageous frame moves it only a bounded amount, and a gentle pull keeps it from sticking: **stiff, but never locked**.

Design only: no agent runs this yet.

source: agents_life_cycle.tex l.1085–1125 (form of the update); lab 3, exercise 5

In the toy: stiff, but not locked



	constant	inferred
sd before (mm)	9.4	6.5
spurious mask (mm)	92	44
after 60 s unseen (mm)	123	45
frames to adapt	9	15

The failure mode: a steady **bias** looks like agreement. A 5 cm bias ends 5.4σ out with constant Q, 7.6σ with inferred.

source: lab 3, exercises 5–6; agents_life_cycle.tex l.1127–1133

Aspect

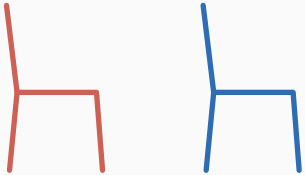
Aspect: the class can be imagined, the instance can be recognised

“How do you imagine a chair?”



the class: what any chair might look like
held inside chair_concept

“Is this the one I saw?”



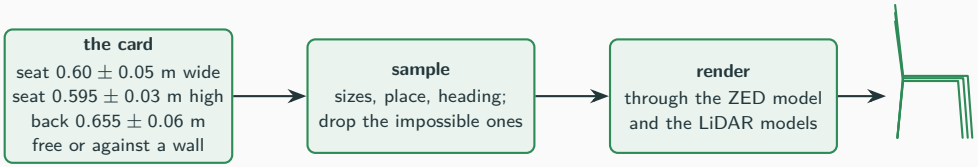
chair_22 chair_23

the instances: what these two looked like, kept with each

The same split as geometry: the class is a prior inside the agent, an instance is a posterior of its own.

Aspect is a dimension of the concept, and it exists as properties of its instances.

How do you imagine a chair? Sample the prior and render it



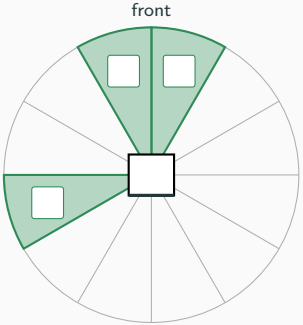
drawn on the chat canvas

- It asks the question "could I see one from there?" of something that is not in the room: the same samples give a probability of detection.
- A chair on the graph is fitted as a pose of a fixed template. Imagination also draws how real chairs vary.
- Today the class's look is a shape, written by hand. No colour, and nothing learned from the chairs it has met.

source: language_agent/src/imagine_view.h; common/concept_manifest/chair.concept.toml

An instance's look is measured one direction at a time

work in progress



12 directions of 30°; the best view of each is kept

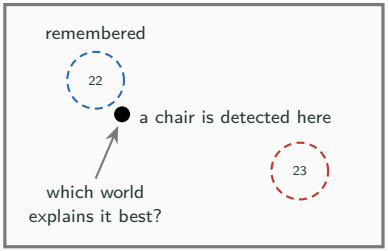
what is kept	where it stands
colour of each instance	runs, for display only; deliberately in no fit
a small crop per direction	on a branch; stored by long-term memory across lives
a learned descriptor per direction (the atlas)	runs in the chair agent in shadow mode; not in long-term memory

A view from the front says little about the back, so a look is a set of views, each tagged with where it was seen from.

Still unreliable: near-identical crops have been filed in far-apart directions.

source: common/appearance_belief; common/view_atlas; branch feat/ltom-thumbnails, 2026-11-09

A second sight is a comparison, and “new” is on the list



is it chair 22?	$\ln p(x \mid 22) + r_{22}$
is it chair 23?	$\ln p(x \mid 23) + r_{23}$
is it **new**?	$\ln(1/A)$

Where it is: the chair stayed, or it was moved with probability π and could be anywhere:

$$p(x \mid k) = (1 - \pi)\,\mathcal{N}(x; \mu_k, \Sigma) + \pi/A.$$

What it looks like: r_k, the nats by which this chair’s views explain the view better than the class does.

Long-term memory only offers. The chair agent decides, the winner must beat “new”, and a tie goes to “new”.

On position alone a moved chair cannot win: $\pi/A < 1/A$.

source: common/recall_match/recall_match.h; common/view_atlas/view_atlas.h

Second sight: what the toy says, what the robot does

work in progress

Toy, % identified correctly

	position	+ look	+ look, same model
chair stayed	98	100	99
chair moved	6	100	33
new chair	81	100	0

With views twice as noisy as the model assumes, the look keeps only 42% of the chairs that never moved. Position alone kept 98%.

Measured: chairs, in simulated tours

Position only, no appearance term. Back in the room, chairs return under their old identity (p .92–.98), also from a previous start of the system.

- The chair is absent for about 35 s after the return, until it is detected again.
- A moved chair is not established: replaying the atlas offline, 15 of 53 right.
- Other classes: merged, never run. Renaming and learned permanence: designed only.

source: lab 3, exercises 18–19; runs ltom_chair2, rose_chair3, 2026-11-08/09

Lab

Lab 3: a concept's life, its footing, its look

`labs/day3.ipynb`, about two hours.

Life cycle

A. **Birth**: ΔF against a streak of three.

B. **Absence**: line of sight through walls; four rules on one tour.

C. **Forgetting**: constant Q against inferred volatility.

Geometry

D. **The room**: the vote, the band, four walls.

E. **The pose**: drive the route; NEES on three maps.

F. **Symmetry**: F at the true pose and its half-turn.

G. **Ceiling**: independent against Woodbury.

Aspect

I. **Second sight**: three remembered chairs and "new"; position against look; one model of chair; a view from the other side.

Short of time: A, B, E, G and I.

Tomorrow

Today we asked what a chair is, how long it lasts and how it looks. Tomorrow: what it is **for**.

- function: objects publish what they can do for the robot, priced in nats;
- expected free energy: reaching a goal and reducing uncertainty in one currency, and how the two mix;
- the controller: routes and tracking as control problems with no separate obstacle term;
- and the same objective turned on the body: calibration as inference.

Reading for Day 4

Thesis chapters:

- *Action: affordances and the expected free energy*, up to "Why the gradient decomposes";
- *Self-Calibration of the Robot's Body*: "What the body is", "What no data can determine", "What each motion makes observable".

Backup: Birth is a comparison between two worlds

$$\Delta F = F^{(n+1)} - F^{(n)} = \underbrace{\mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n+1})] - \mathbb{E}_q[-\ln p(\bar{Z} \mid \theta_{1:n})]}_{\text{accuracy gained by explaining those points}} + \underbrace{D_{\mathrm{KL}}[q(\theta_{n+1}) \,\|\, p(\theta_{n+1} \mid \ell)]}_{\text{complexity paid for the new object}}$$



born exactly when the left side outweighs the right: $\Delta F < 0$

Two models of *different order*, one currency. The complexity is not a penalty added for regularisation: it is the same term every live instance already pays.

source: agents_life_cycle.tex l.594–622

Backup: The room is the first belief born

$$F(m) = \underbrace{\mathbb{E}_q[-\ln p(o \mid X, \theta_m, m)]}_{\text{inaccuracy}} + \underbrace{\kappa\,|V|}_{\text{complexity}}, \quad \kappa = \ln \frac{L}{\sigma}$$

The number of walls is unknown, so the room is a choice of *structure* m, priced like a birth. One coordinate to $\sigma = 3$ cm in a room 10 m across costs

$$\ln(10/0.03) \approx 5.8 \text{ nats.}$$

Minimum description length and minimum free energy are, here, one rule.



pays if its cells are worth more than its corners

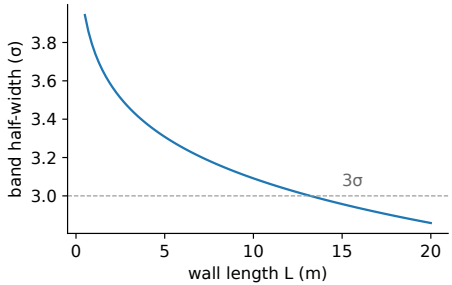
source: layout_acquisition.tex l.56–66, l.128–146

Backup: A band that is derived, not chosen

Clutter is uniform over A. A wall puts its points in a strip L long, σ wide:

$$\frac{p_{\text{wall}}}{p_{\text{clutter}}} = \frac{A}{L\sqrt{2\pi}\,\sigma}\, e^{-r^2/2\sigma^2}$$

- Inlier iff the ratio exceeds 1: about 3σ.
- Each point brings ln(ratio) **nats**.
- The band **narrows as L grows**: a long wall is a more expensive explanation.



A tuned threshold gives the band. The derivation gives the band *and* the currency.

source: lab 3, exercise 8; the same mixture, per cell: layout_acquisition.tex l.1049–1055

Backup: Promotion: the outline with the shortest description



schematic

The outline moves between wall and furniture as the robot drives, so "the latest one" would make the room depend on when exploration stopped. Up to 60 recent outlines are kept and scored on *all* the evidence.

- In 7 of 10 apartment runs (earlier build) the winner was not the last drawn.
- From the kitchen-corner start: overlap 0.968 at promotion, 0.983 refined.
- Ten generated rooms: median 0.994.

All in simulation; the pipeline has not run on the physical robot.

source: layout_acquisition.tex l.1036–1046, l.1173–1177, l.1455–1462

Backup: Observed, precision stops at a floor

One predict plus one update, $\Sigma^{-1} \leftarrow \Sigma^{-1} + i$, with y the predicted precision and $\varrho = Q$:

$$\frac{1}{y} = \frac{1}{y+i} + \varrho \quad \Longrightarrow \quad \boxed{\sigma_{\text{steady}} \approx (\varrho/i)^{1/4}}$$

- Elapsed time does not appear. Evidence is discarded at the rate it is acquired: three hours of observation leave the belief where thirty seconds did.
- Lab: formula 10.0 mm, simulated 9.4 mm.
- Clamping Σ caps the first problem and does nothing about this one.

Nothing an agent learns extends how long it stays resistant.

source: agents_life_cycle.tex l.1045–1082; lab 3, exercise 5

Backup: A constant becomes an inferred quantity

Each instance carries a log-volatility, $Q = e^{\zeta}$, driven by its own surprise:

$$\zeta \leftarrow \zeta + \eta\Big(\underbrace{\tfrac{1}{2}\,\tilde{d} - \tfrac{1}{2}\mathbb{E}[\tilde{d}\,]}_{\text{surprise seen minus expected}} - \underbrace{\frac{\zeta - \zeta_0}{\sigma_\zeta^2}}_{\text{hyperprior pull-back}} \Big)$$

- Consistent views: ζ falls, memory lengthens, precision accumulates.
- A sustained surprise: ζ rises and the model adapts. Stiffness is never a lock.

In the lab $\tilde{d}$ is the saturating (Student-t) surprise of the innovation: one outrageous frame moves ζ by a bounded amount.

Design only: no agent runs this yet.

source: agents_life_cycle.tex l.1085–1125 (form of the update); lab 3, exercise 5

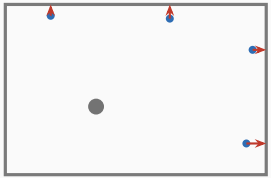
Backup: The room supplies the likelihood

For a scan $\{p_i\}$ and a candidate pose X:

$$d_i(X) = \text{dist}\left(R(\phi)p_i + t,\ \mathcal{R}\right)$$

$$\mathcal{L}_{\text{obs}} = \frac{1}{2\sigma^2} \cdot \frac{1}{N} \sum_i w_i \,\text{Huber}(d_i)$$

- The residual *is* the prediction error: no expected scan is rendered.
- Note the $\frac{1}{N}$: the precision does not grow with scan density.
- Range and incidence weights are precisions on a covariate, not gates.



each return is pulled onto its nearest wall

source: room_concept.tex l.243–307

Backup: What the residual floor is made of



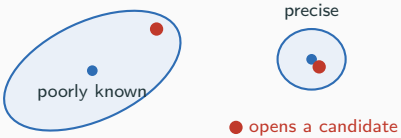
Park the robot. The mean wall distance of each scan hovers near 87 mm and never reaches zero.

- Its fluctuation is white: $N^{-1/2}$, measured ratio 1.41 against a predicted $\sqrt{405/200} = 1.42$.
- That fixes the per-point scatter at ≈ 37 mm, which explains $0.798 \times 37 \approx 30$ mm.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.386–426

Backup: Association: the factor the free energy cannot see



a poorly known instance accepts detections from further away

$$P(M \mid X, Z, \Phi) \approx \mathbf{1}[M = \hat{M}], \quad S = \Sigma^{xy}_{\theta_i} + \sigma^2_{\text{det}} I$$

Squared Mahalanobis distance at most 9, assigned greedily, lowest cost first.

A distribution on one value has no entropy and no expectation: the matching is free, in the literal sense that F cannot see it. A wrong match is invisible to the quantity that would have registered it.

The repair comes later and from elsewhere: merging, and the existence belief.

source: concept_agents.tex l.483–486, l.859–889

Backup: The common-mode ceiling

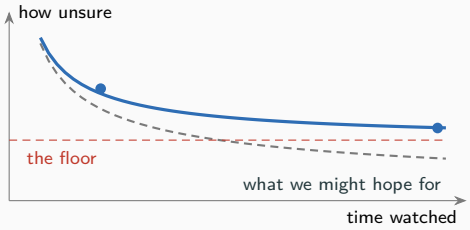
N residuals that share one offset $\delta \sim \mathcal{N}(0, \Sigma_\delta)$, a pose error, plus independent noise. Marginalise δ:

$$\mathcal{I} = J_z^\top \left(\sigma^2 I + J_z \Sigma_\delta J_z^\top\right)^{-1} J_z \overset{\text{Woodbury}}{=} \left(\sigma^2 (J_z^\top J_z)^{-1} + \Sigma_\delta\right)^{-1} \xrightarrow[N\to\infty]{} \Sigma_\delta^{-1}$$

- A denser mask eventually stops helping. The ceiling is set by the *pose* error, not by the pixel count.
- The joint information on (object, offset) has rank 6, not 12: the data cannot tell a displaced object from a displaced robot.

source: concept_agents.tex l.1048–1090

Backup: Precision stops at a floor, in a picture



The doubt added each frame is balanced by what that frame teaches, and the belief stops improving.

Elapsed time does not appear. Three hours of observation leave the belief where thirty seconds did.

Lab: 10.0 mm predicted, 9.4 mm simulated.

Nothing an agent learns extends how long it stays resistant.

source: agents_life_cycle.tex l.1045–1082; lab 3, exercise 5

Backup: The residual floor, in a picture



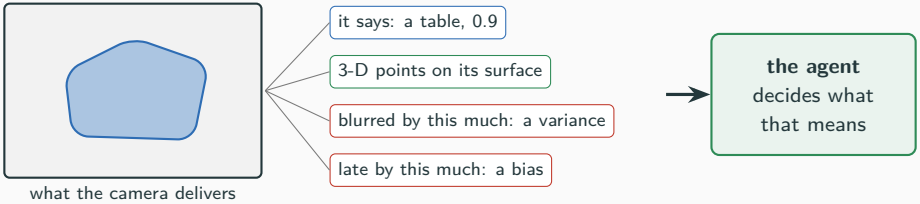
Park the robot. The average distance from its scan to the walls sits near 87 mm and never reaches zero.

About 30 mm is sensor noise. The other 57 mm is where the map does not match the world, and it is the same on every scan.

Two-thirds of the floor is an error every return shares. More returns cannot remove it.

source: room_concept.tex l.386–426

Backup: A mask arrives with a label saying how far to trust it



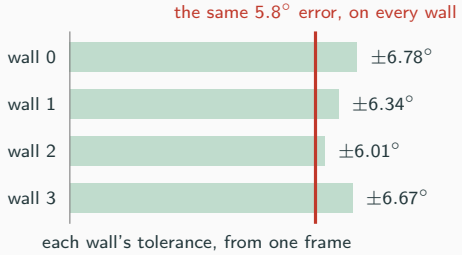
The producer reports the physics of the capture and gates nothing; the agent decides what that does to its belief.

Variance and bias are different things, and they are reported apart: an error that averages lowers the weight; an error that does not stops the update.

source: phylogenetic_perception.tex l.347–408

Backup: Case study: association dies by a common-mode latch

work in progress



This is one **heading** error, not four wall errors, yet each wall is tested on its own against it.

When it crosses about 6°, they all fail in the same frame. Nothing is left to correct the pose, and association goes from 100% to 0% for ever.

"A parked run looks perfect and the first tour kills it."

source: an earlier estimator; post-thesis note, 2026-09-18